

JAIVARDHAN SINGH

Jamshedpur, Jharkhand, India | +91 8709938626 | jaivardhan.workmail@gmail.com

[LinkedIn](#) | [GitHub](#)

SUMMARY

Computer Science undergraduate at VIT Vellore (CGPA: 8.16/10). Built production automation tools in VB.NET and Oracle SQL during software development internships at Timken India and Tata Cummins, deployed on live manufacturing floors. Separately ships full-stack web projects in Next.js and MongoDB, including recent work on transactional concurrency control and atomic booking logic. 200+ DSA problems solved in C++, backed by coursework in DBMS, Operating Systems, OOP, and Computer Networks. Seeking SDE roles.

EDUCATION

Vellore Institute of Technology, Vellore — B.Tech, Computer Science Engineering (CGPA: 8.16/10) *Aug 2023 – May 2027*
Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming

Vidya Bharati Chinmaya Vidyalaya, Jamshedpur — Class XII: 80.0% *2021 – 2023*

Gulmohur High School, Jamshedpur — Class X: 81.5% *2008 – 2021*

EXPERIENCE

Software Developer Intern — Timken India Limited *May 2025 – Jun 2025*

- Engineered a Network Device Management System in VB.NET and Oracle SQL that centralised IT asset tracking for 200+ network devices across multiple company locations, replacing scattered spreadsheet-based records
- Designed normalised master data entry modules (Company, Location, Asset Type) that standardised and automated asset registration workflows, cutting manual data entry time by approximately 35%
- Implemented real-time device monitoring with automated email alerting, reducing downtime detection on critical network infrastructure from 15+ minutes to under 1 minute

Software Developer Intern — Tata Cummins Private Limited *Dec 2024 – Jan 2025*

- Developed end-to-end quality inspection automation software in VB.NET and Oracle SQL, replacing manual paper-based checks across the production line and improving inspection throughput while reducing human error
- Integrated camera-based verification and QR code generation into the inspection flow, strengthening part traceability and cutting manual verification steps by approximately 50%
- Built a data access and audit module enabling engineers to retrieve historical inspection records for defect analysis and compliance reporting

PROJECTS

Bookify – Concurrent Ticket Booking Platform | [GitHub](#) | [Live](#)

- Built a full-stack movie and concert ticket booking platform on Next.js 16 (App Router) with Route Handlers as the backend API, a Mongoose/MongoDB data layer, and JWT httpOnly-cookie auth across three roles (customer, organiser, admin)
- Eliminated double-booking by modelling every seat as an independent document and acquiring it through a single atomic conditional write (available → held), so two simultaneous requests for the same seat can never both succeed
- Wrapped multi-seat holds, booking confirmation, cancellation, and waitlist acceptance in MongoDB multi-document transactions, and enforced TTL-based seat holds both lazily on every read/write and via a scheduled cron sweep
- Implemented a FIFO waitlist that atomically cascades cancelled seats to the next customer as a time-limited emailed offer, with QR-coded e-tickets over Nodemailer and an organiser revenue dashboard; deployed on Vercel with MongoDB Atlas

InterviewAI – AI Voice Interview Simulator | [GitHub](#) | [Live](#)

- Built a voice-driven mock interview simulator with hold-to-speak capture, live speech-to-text, and spoken AI feedback across 6 technical roles (SDE, Frontend, PM, DevOps, Data Analyst, System Design) using Next.js and the Web Speech API
- Integrated the Google Gemini API with automatic model discovery for real-time answer evaluation, per-question scoring, and an end-of-session performance summary
- Engineered retry-with-backoff and model-fallback logic to absorb Gemini rate limits mid-interview, ensuring a session never drops a candidate response

Placement Tracker System – Multi-Role Placement Platform | [GitHub](#) | [Live](#)

- Developed a multi-role placement portal with company listings, application tracking, deadline management, and real-time statistics dashboards for both students and administrators
- Designed RESTful APIs over MongoDB Atlas with JWT cookie-based authentication and role-specific responsive UI; deployed on Vercel with CI/CD on every push to main
- Handled token expiry and silent refresh through httpOnly cookies, preventing abrupt session drops and unnecessary re-logins during active dashboard use

TECHNICAL SKILLS

- Languages: C++, JavaScript, SQL, Java, VB.NET
- Frontend: React.js, Next.js (App Router), Tailwind CSS, HTML, CSS
- Backend: Next.js Route Handlers, REST APIs, JWT Authentication, Mongoose, Nodemailer
- Databases: MongoDB, MongoDB Atlas, Oracle SQL
- Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks
- Tools: Git, GitHub, Docker, Vercel, VS Code

ACHIEVEMENTS

- 3rd Place, IEEE SENSE-A-Thon 2026** – Built a real-time Vehicle Health Monitoring System with multi-sensor integration, placing 3rd against 50+ engineering college teams
- 200+ DSA Problems Solved** – Solved 200+ problems in C++ across LeetCode and GeeksforGeeks, covering Arrays, Strings, Graphs, Trees, and Dynamic Programming